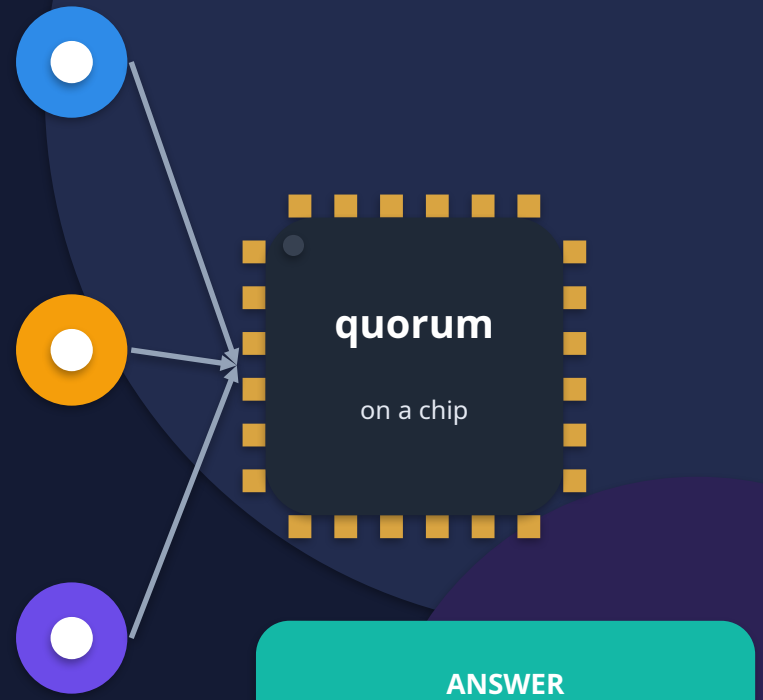
BIODREAM LLC · FINAL OVERVIEW

quorum

A tiny, honest judge for machines.

It reads a machine's sensors, weighs the evidence like a courtroom, gives one clear answer with its reasons, and was checked against every situation before it was allowed to run.



Plain-English rules | **Checks every case** | **Explains itself** | **Runs on an $11 chip**

What it is · what it does · the problems it solves · where else it fits · where it stands

AT A GLANCE

quorum in six pictures

The whole idea on one page. Each of these gets its own slide.



Rules in plain words

Engineers write what they already know. No programming.



Rules argue

Like a courtroom: evidence, excuses and doubts knock each other down.



Every case checked

Every possible combination of readings is tried before it runs.



Tiny and on time

Fits an $11 chip and always answers within a proven time.



Says why

Every answer lists what it relied on, and what it had to assume.



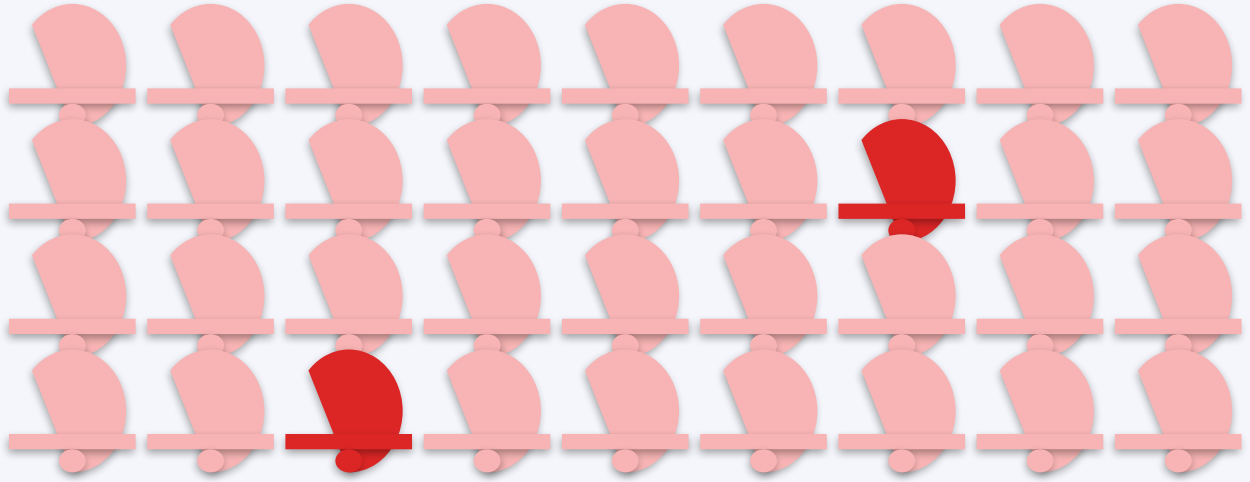
Refuses bad input

A rulebook with a hole, or a damaged file, is rejected outright.

THE PROBLEM

Machines cry wolf, and people stop listening

Plants, buildings and utilities are full of alarms. Most are noise, and none of them explain themselves.



Pale: nuisance alarms. Dark: the two that mattered.

Alarm fatigue

When most alarms are wrong, people learn to ignore them, including the one that is right.

No reasons given

"The model scored 0.83" tells an operator nothing they can check or argue with.

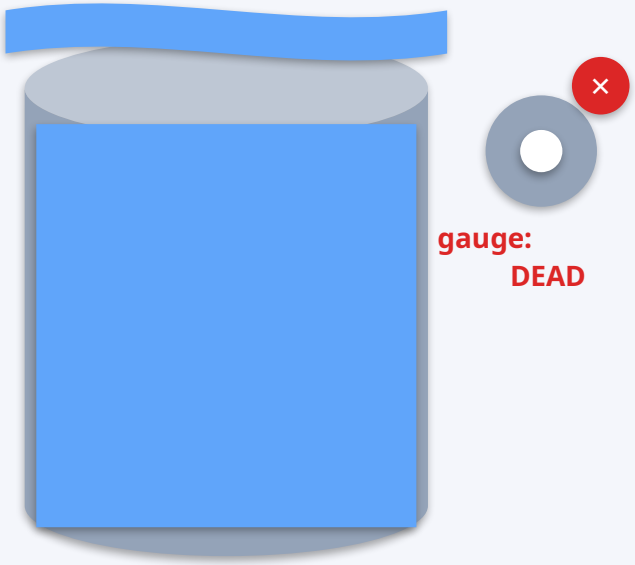
What people actually need

One clear answer, the reasons behind it, and an honest note of anything it could not see. That is what quorum is built to give.

THE TRAP NOBODY SEES

A dead sensor looks exactly like a healthy one

Silence from a broken gauge reads as "all is well". Most systems cannot tell the difference.



ORDINARY SYSTEM
"No high-level signal. Tank is fine."

QUORUM
"I am ASSUMING the level is not high, because the gauge said nothing. Check it."

Real history

The 2009 CAPECO tank-farm explosion in Puerto Rico began with a tank overfilling while its level gauge was not working. quorum names what it assumed, so a dead gauge shows up in the explanation.

quorum also checks the rulebook itself: it warns when nothing in it can tell a dead sensor from a quiet one, before it ever goes near a machine.

IN ONE SENTENCE

What quorum is

A rulebook for a machine that is allowed to argue with itself, checked against every situation that could ever happen, small enough to run on a chip that costs less than lunch.

1 Write

An engineer writes rules in plain words, or imports a safety worksheet.



the rulebook

2 Check

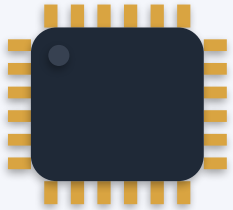
The computer tries every combination of readings and refuses if any fails.

✓

every case passed

3 Load

One click turns the checked rulebook into a file for the chip.



loaded on a chip

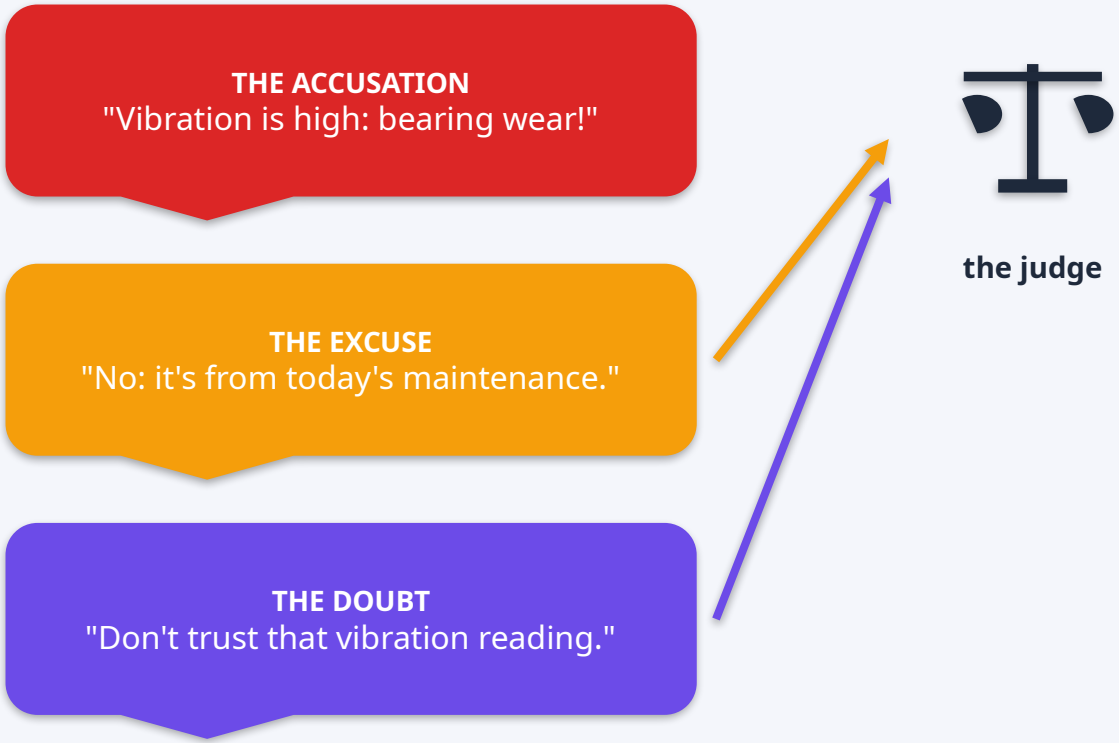
4 Decide

The chip answers in well under a millisecond, with its reasons.

CHIP
"Carry on: maintenance today."

Like a courtroom, not a calculator

Every rule is an argument. Arguments can knock each other down. Whatever survives decides.



How a case plays out

Vibration alone: the accusation stands. Stop the pump.

Vibration + maintenance today: the excuse knocks it down. Carry on.

Vibration + loose sensor mount: the reading itself is in doubt. Fix the sensor first.

Exactly one answer every time, and the reason is the chain of arguments that survived.

This style of reasoning ("argumentation") has decades of research behind it. quorum's contribution is making it small, checked and on time enough for real machines.

Engineers write it in plain words

A short, readable file. An expert can read it, argue with it and sign it.

```
casebook    centrifugal-pump
observe     vibration_high, bearing_temp_high, ...
argument    bearing_wear  "wear in the bearing"
attacks     maintenance_today -> bearing_wear
action      stop_pump  when bearing_wear
safe        stop_pump
source      FMEA-PUMP-001 rev C, J. Okafor
```

No programming

A handful of keywords. The computer writes all the fiddly wiring for the chip itself.

Or start from paperwork

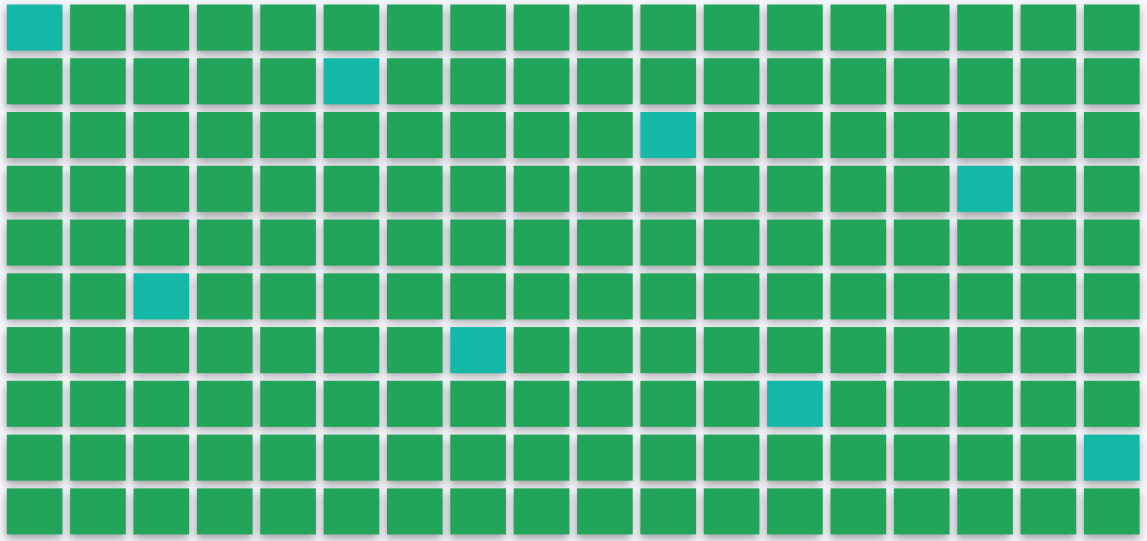
Import the safety worksheets a plant already keeps (HAZOP, LOPA, FMEA) and get a draft rulebook to finish.

Every rulebook carries its source: which document, which revision, who wrote it. The chip reports it back, so you always know which rules it is running.

THE SUPERPOWER

It checks every situation that could ever happen

Not the scenarios someone thought of. All of them, before anything goes near a machine.



Each square stands for combinations of sensor readings.
The pump rulebook has 14 sensors: 16,384 combinations, every one solved and checked.

16,384

pump situations, all checked

491,520

situations checked for two rulebooks sharing one chip

Why this is unusual

Testing finds the problems you looked for. Trying every case finds the ones nobody imagined.

The check is the price of admission: no rulebook reaches a chip without passing all of it.

IT SAYS NO

If anything is wrong, it refuses to build

A safety tool that ships with a hole in it is worse than none. Refusing is the point.



No chip file is written. Nothing reaches the machine.

≠ Two answers at once

A situation where the rules say both "stop" and "carry on".

∅ No answer at all

A situation where every option is knocked down.

⏱ Too slow

Any situation that would take longer than the proven limit.

⊘ A dead action

A response the rulebook names but can never actually choose.

And it says why

Each refusal names the exact situation and the rules involved, so the author can fix the rulebook rather than guess.

THE SURPRISE

It finds holes in safety paperwork people already signed

Plants keep long safety studies. quorum reads them and checks them the way it checks a rulebook.



HAZOP · LOPA · FMEA worksheets

quorum checks

"Independent" protections that aren't

Safety rules say backups must not share a weak point. quorum checks whether every route to a safety action really is separate.

It found a real one, first time

In our own pump rulebook, emergency shutdown only worked if BOTH of two sensors spoke up.

This could be the first business

Consultants charge thousands a day to review these by hand. A list of gaps needs no hardware and no certification.

ALWAYS ON TIME

A promise about speed, proven before it runs

For a machine, a late answer is a wrong answer. quorum works out its worst case in advance.

Pump rulebook: the most it can ever need

248 steps

Pump rulebook: the most it ever took

140 steps

On both chips, every situation finished inside the proven limit.



156 µs

slowest pump decision, Pico 2

350 µs

same on the slower Uno R4

∞ Why smarter systems often can't promise this

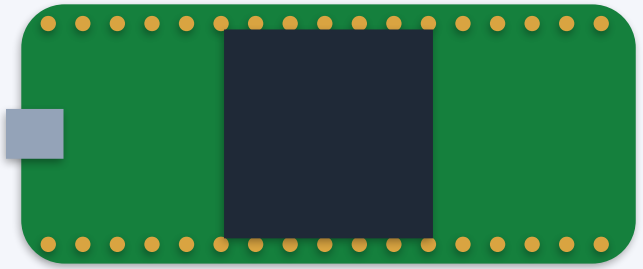
Reasoning by search can, in bad cases, take practically forever. quorum's never can.

µs = millionths of a second. A blink of an eye is about 100,000 µs.

TINY HARDWARE

Runs on chips that cost less than lunch

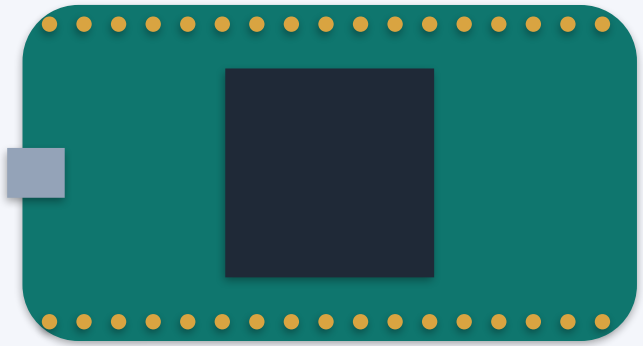
No cloud, no internet, no graphics card. Tested on three real boards from three different makers.



Raspberry Pi Pico 2

About $11 · 150 MHz

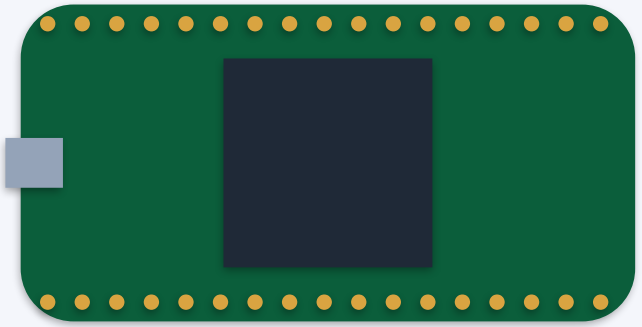
Same answers as the PC in every situation



Arduino Uno R4 Minima

About $20 · 48 MHz

Same answers as the PC in every situation



Raspberry Pi 5

The "bigger computer" · runs the second opinion and the full test suite natively

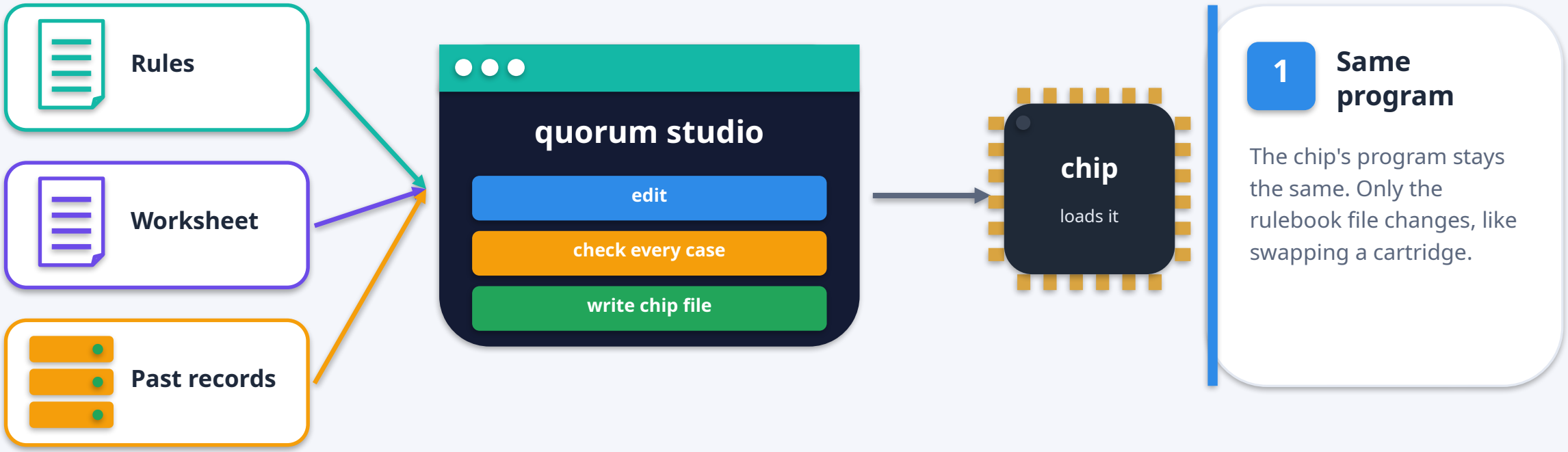
Same answer everywhere

The same rulebook was run on the PC, the Pico 2 and the Uno R4. Every situation got the same answer and took the same number of steps on all three. That is evidence the answers come from quorum, not from a quirk of one chip.

On the Pi 5, all 73 of quorum's automatic tests pass, built from scratch on the board itself.

One chip program. Any rulebook. One click.

A desktop app turns rules, worksheets or past records into a checked file the chip loads.



1 Same program

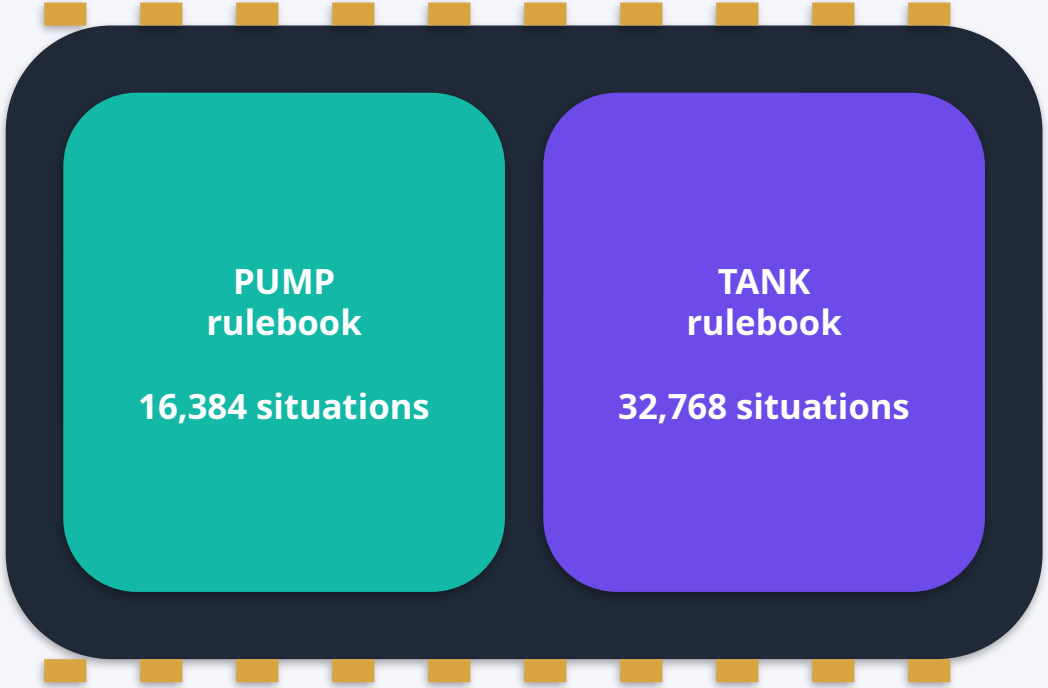
The chip's program stays the same. Only the rulebook file changes, like swapping a cartridge.

Runs on Windows or Linux, installs without administrator rights, and works entirely offline. It writes files for both the Pico 2 and the Uno R4.

ROOM FOR MORE

Several rulebooks can share one chip

A pump and its tank on one board, each still deciding exactly as it would on its own.



✓ Checked for interference

Before they share a chip, quorum proves neither can change the other's answers.

≡ Proved on real chips

On both the Pico 2 and the Uno R4, each rulebook gave exactly its stand-alone answers in every situation.

Up to four rulebooks fit on one chip. One small board can watch a whole skid of equipment.

CAN'T BE FOOLED BY A BAD FILE

A damaged or altered rulebook file is refused

Before it decides anything, the chip checks the file three ways. Any doubt and it stays silent.



Three checks at start-up

1 Arrived intact?

A checksum catches a single flipped bit.

2 Makes sense?

Every count and cross-reference is in range.

3 The right rules?

A fingerprint of the rules must match the one the checker signed off.

Tried for real, on the chips

One flipped bit: refused. Two rules swapped, checksum faked: refused. A bearing alarm limit quietly raised from 90 to 95 °C, checksum faked: refused (Uno R4). Each time the chip made no decision at all.

A machine with no trustworthy rulebook must not act on one. quorum fails safe.

Every decision comes with its reasons

Not a score. A short, readable account an operator can check, and argue with.

DECISION: stop the pump

Because: bearing wear is the explanation that survived.

Relied on: vibration high, bearing temperature high.

Ruled out: structural resonance (bearing temperature contradicts it).

Assumed absent: loose mount, overdue calibration.

No sensor reported those, so quorum says it assumed them.

Why that last line matters

Other systems treat "no news" as good news, silently. quorum writes the assumption down.

Built for audits

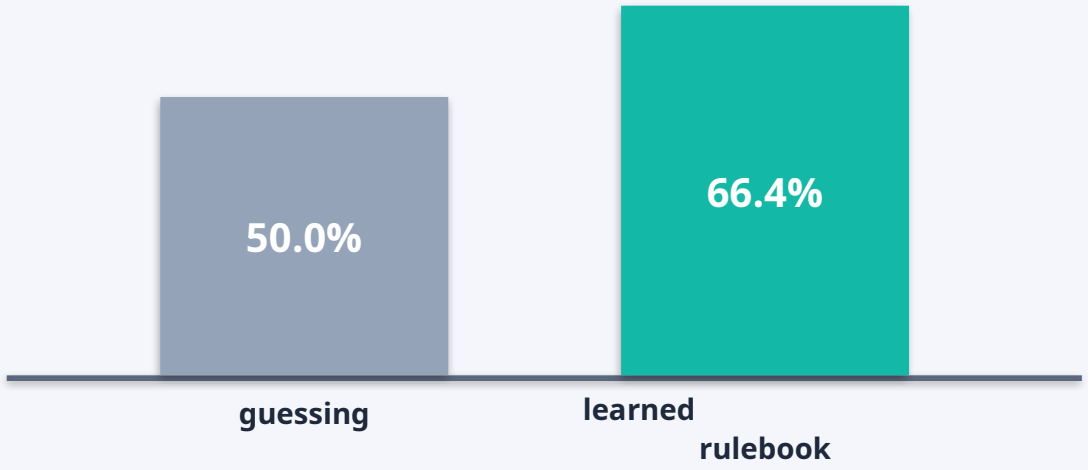
Which rulebook, which revision, which facts, which rules. Enough to answer "why did it do that?" later.

If a sensor is dead, the explanation shows the decision rested on something nobody actually measured.

It can draft a rulebook from past records

Give it a maintenance or incident history. It writes rules in the same plain language, for an expert to approve.

Spotting machine failures in a public dataset of 10,000 records (AI4I)



⚠ It catches cheating columns

It warns when a column is only filled in after a failure, which would make it look 98.5% accurate and useless in real life.

≈ What we can honestly say

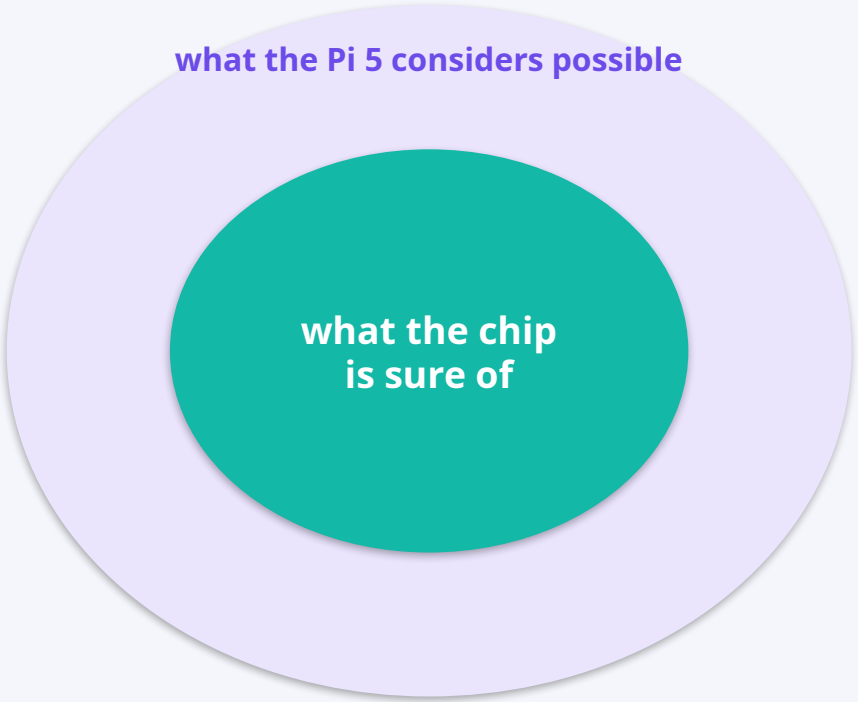
In four tests set up in advance, learning added 8 to 20 points on average, but never cleared every bar.

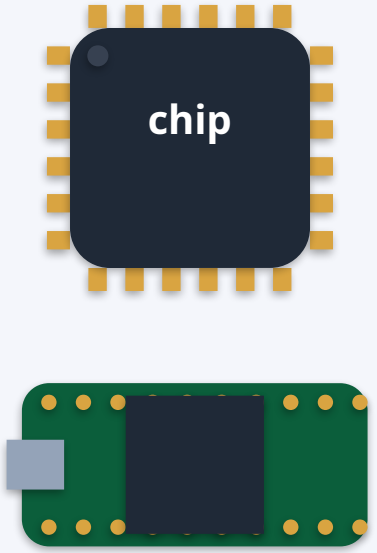
So: drafts, not decisions

Learning happens on a computer and only produces a draft. A person approves it, and it must pass the same every-case check.

The chip decides. A bigger computer double-checks.

And mathematically, the two can never contradict each other.





The chip: fast and certain

Gives the one cautious answer, on time, every time, on the machine itself.

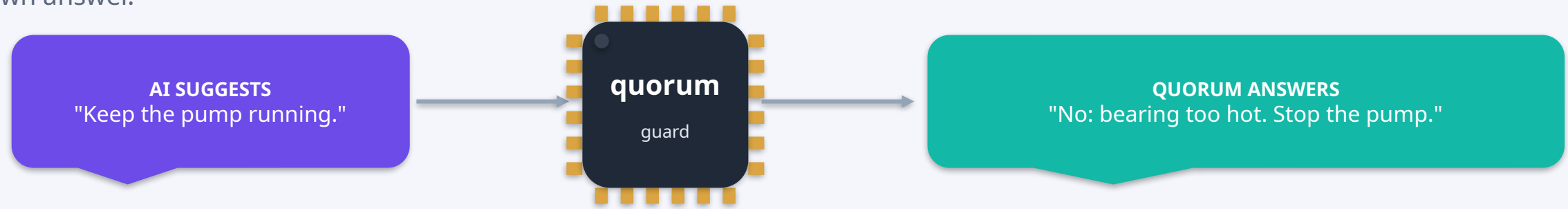
The Pi 5: every other reading

Lists every other reasonable way to read the same facts, for a person to look at. It only advises.

Checked: across every situation of three rulebooks, 50,432 comparisons, the bigger computer never disagreed with anything the chip was sure of.

Another system can propose; quorum has the last word

An AI model or an ordinary controller suggests an action. quorum lets it through only if it is at least as cautious as the rulebook's own answer.



✓ Checked every pairing

Every situation times every possible suggestion: 49,152 for the pump, 131,072 for the tank. Never once less cautious than the rulebook.

⚙ Same on the chips

The same result was reproduced on the Uno R4 and the Pico 2, not just on the PC.

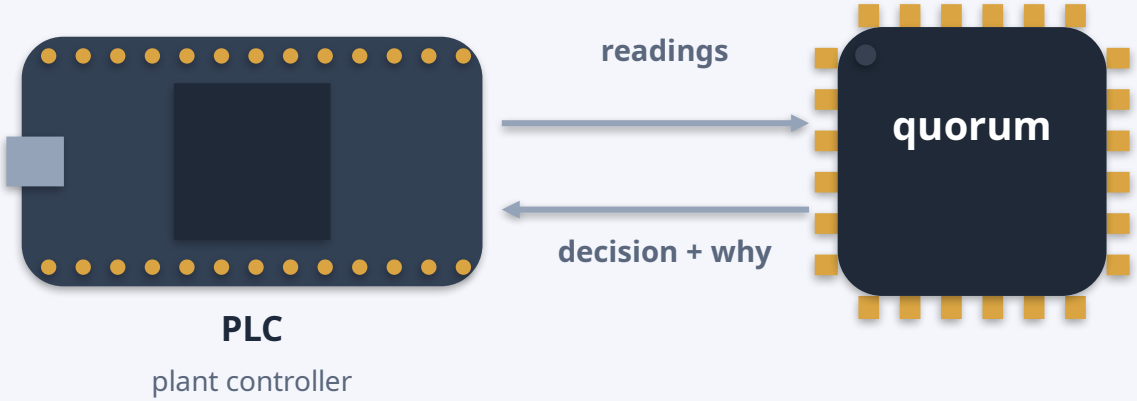
! Only as good as the rulebook

It enforces the rulebook a person approved, including its ranking of actions. A wrong rulebook is enforced too.

This is a possibility for AI in control rooms, not yet proven in the field.

It speaks Modbus, the language most factory controllers use

A PLC writes sensor values into quorum's registers and reads back the decision and the reason.



= Same answer either way

All 16,384 pump situations, sent through the registers, got exactly the decision the chip makes directly. On the Uno R4 and the Pico 2.

× Ignores garbage

A damaged message is dropped. A command it does not know is refused. It changes nothing unless a request is valid.

Important limit

Modbus has no password. Anyone on the same wire can send it readings. It must sit on a closed, trusted network, as Modbus devices always should.

PROOF YOU CAN CHECK YOURSELF

Every situation checked at once, with a receipt

No need to trust quorum's own checker. It hands over a proof that a separate small program verifies.

All at once

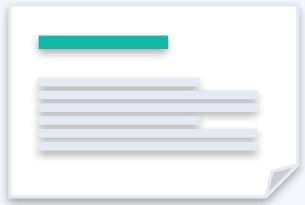
A second checker reasons about every situation together. Two rulebooks sharing one chip: all 536,870,912 combinations, with no rulebook affecting the other.

A certificate

A file listing, for every situation, what quorum decided and why. Tied to the exact rulebook by its fingerprint.

An independent check

About 200 lines of separate code re-check the certificate. Every forgery we tried was caught: a changed answer, a moved rule, a nudged alarm limit.



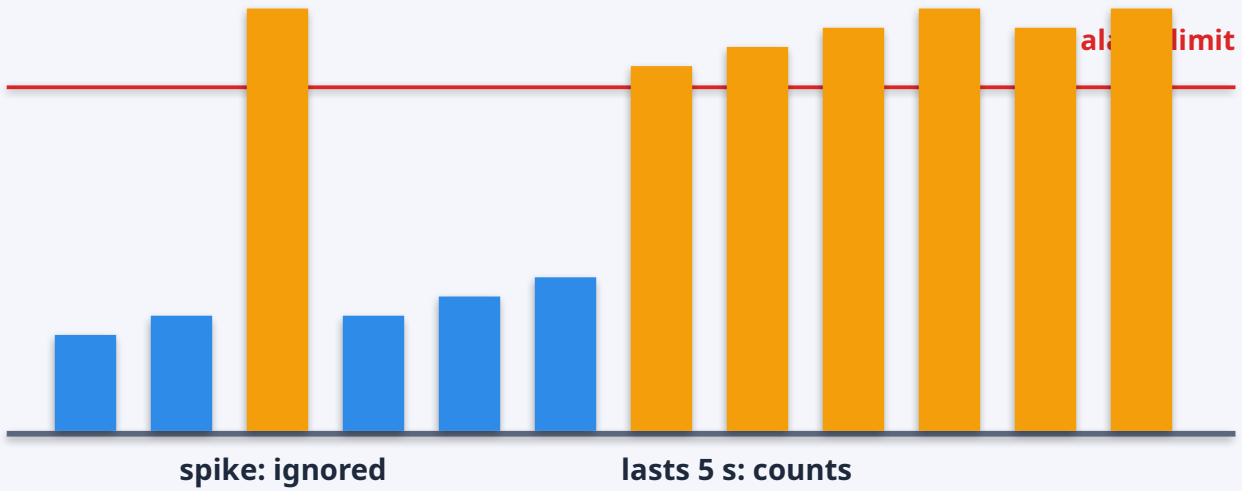
CERT

An auditor can confirm the rulebook was checked without trusting us or our software.

IT WAITS BEFORE IT ACTS

"Hot for 5 seconds", not "hot for a moment"

A spike should not stop a plant. Rules can say a reading must last before it counts.



Waits only in the safe direction

Turning an alarm ON waits. Turning it OFF never does, because delaying that is the dangerous way.

Still every case checked

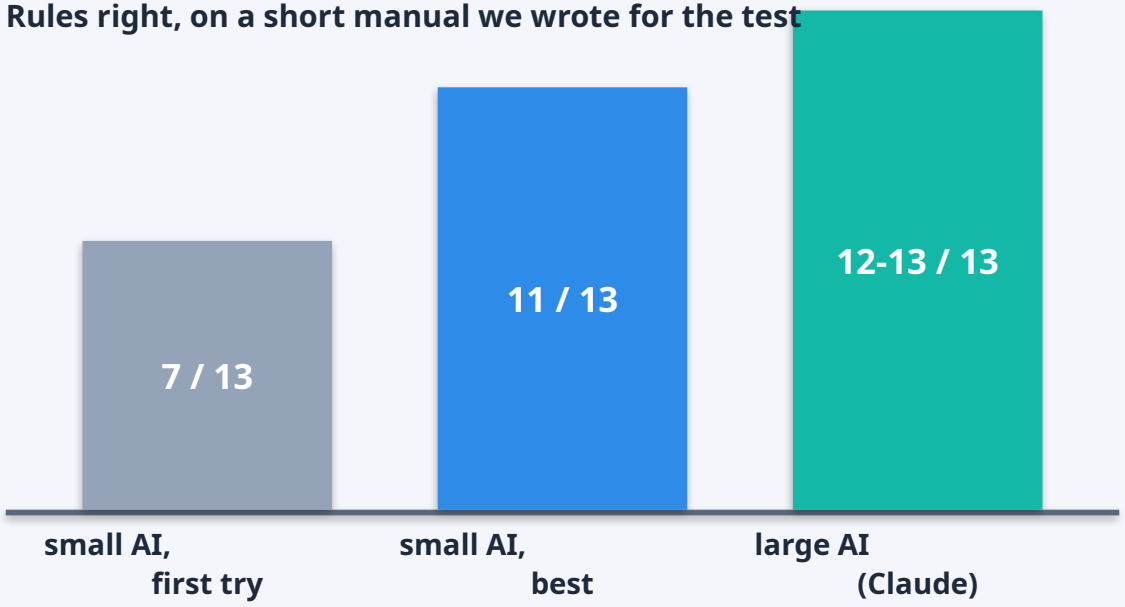
Time only decides when a reading becomes true, so the every-situation check stays complete.

This is how alarm standards are already written (ISA-18.2), now in plain words.

An AI can write a first draft. A person must approve it.

Tried with a small AI on our own computer, and with large ones through a service that keeps no copy.

Rules right, on a short manual we wrote for the test



≈ On real documents: getting there

Four regulations it had never seen, answers written down first: 86% of checks right, 5 runs each.

! The worst mistake sounded reasonable

One draft let a power blip cancel real shutdowns. The review flagged it; only a person can judge it.

So: a helper, not an author

Results still vary from run to run. Every rule must quote the manual, and an unapproved draft can never reach a chip.

WHY THIS MATTERS

AI-drafted, machine-checked, engineer-approved

AI is fast but makes mistakes. quorum's checks catch most of them. A person makes the final call.

1



AI drafts

Reads the manual and writes every rule, each with the exact sentence it came from. Minutes, not weeks.

2



quorum checks

Throws out any rule that misquotes the manual, sends every refusal back to the AI to fix, and checks every possible situation.

3



An engineer approves

Reads a short list of flagged items instead of writing from scratch. Nothing reaches a chip without this.

✓

What the checks guarantee

Every rule traces to the manual, every situation gets exactly one answer, nothing contradicts itself.

!

What they cannot guarantee

That a rule means what the manual meant. A tidy but wrong rule still needs the engineer.

The AI never has the last word. That is what lets it near a machine.

What other approaches can't do all at once

A fair comparison: each alternative does some of this well. Doing all of it together is the unusual part.

	Simple alarm thresholds	AI / machine-learning model	Classic rules engine	quorum
Says why it decided	~	×	✓	✓
Says what it had to assume	×	×	×	✓
Checked against every case first	×	×	×	✓
Proven time limit	✓	✓	×	✓
Fits a very small chip	✓	~	×	✓
Learns from data	×	✓	×	~

✓ yes ~ partly × not usually. quorum's learning is "partly" because it only drafts rules for a person to approve. Small AI models can fit on chips, but cannot be checked case by case.

Anywhere a few sensors drive a decision

The same idea fits many everyday places. These are possibilities, not yet proven in the field.



Buildings

Boilers, heating and cooling, sump pumps: "run, shut down, or call someone" with the reason.



Farms

Irrigation pumps, barn ventilation and grain dryers, often far from help and the internet.



Water and sewer

A town's lift stations and pumps: catch the dead float switch before the overflow.



Labs and cold storage

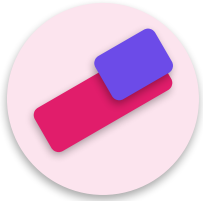
Freezers holding samples or vaccines: a door, a compressor and a thermometer that can lie.

What these have in common

A handful of sensors, a few possible actions, a cost to getting it wrong, and a person who will later ask "why did it do that?". That is exactly the shape of problem quorum was built for.

It also works where the "machine" is a set of rules

quorum checks any rulebook with exceptions. It does not have to run on a chip at all.



Hospital pharmacy

Check the rules that flag patients for a pharmacist to review, with their exceptions.



Batteries and solar

Home batteries, EV chargers and solar inverters: charge, hold or disconnect, with reasons.



Boats and RVs

Bilge pumps, generators and fridges on boats and RVs, where a silent sensor is common.



Computer systems

IT and data-centre alerts: cut the alarm flood and say which alert actually matters.



Business rules

Insurance, lending or compliance checklists: find the case no rule covers, before a customer does.



Fleets and logistics

Refrigerated trucks and delivery fleets: one small box per vehicle, no signal needed.

Machines people trust with their lives

Where a dead sensor must never look like "all clear". Possibilities, not yet proven in the field.



Elevators and escalators

Door, brake and overspeed sensors: take it out of service, and tell the technician exactly why.



Railway level crossings

Gates, lights and train detectors: a silent detector must never read as "no train coming".



Mines and road tunnels

Gas, airflow and fan sensors: when to ventilate, when to evacuate, and when the gas sensor is lying.



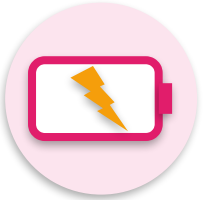
Theme-park rides

Restraints, brakes and position sensors: the ride only starts when every check has positively passed.



Fire alarms in buildings

Check the cause-and-effect rules: every alarm zone opens the right doors and starts the right fans.



Hospital backup power

Generator start, fuel level and transfer switch: is the hospital really ready for a blackout?

Food, farms and living things

Where a wrong call costs a harvest, a batch or a herd. Possibilities, not yet proven in the field.



Fish farms

Oxygen, temperature and aerators: a stuck oxygen probe can lose a whole pen overnight.



Greenhouses and vertical farms

Heat, humidity, CO2 and watering: act on the plants, not on a sensor that has dried out.



Breweries and wineries

Fermentation temperature and pressure: cool, vent or hold the tank, with a record for every batch.



Dairies and milking

Milk cooling, wash cycles and tank temperature: never ship a batch after a missed wash.



Grain silos

Temperature cables, moisture and fans: catch spoiling grain early, and the dust risk before it builds.



Food-safety plans

Check a kitchen or factory's food-safety rules: every hazard has a control, nothing contradicts.

Everyday services and rulebooks

Where people need a clear answer and a reason. Possibilities, not yet proven in the field.



Pools and water parks

Chlorine and acid dosing: never dose when the flow sensor is dead or the pump has stopped.



Wind and solar farms

Icing, vibration and grid faults: keep turning, slow down or stop, with the reason for the crew.



Electric-vehicle chargers

Cable heat, ground faults and battery limits: charge, slow down or stop, explained.



Aircraft and fleet upkeep

Maintenance checklists with exceptions: find the case where two rules disagree, before a mechanic does.



Benefits and grants

Eligibility rules with exceptions: a missing document is "not yet known", never quietly "no".



Smart homes

Leak sensors and water shut-off valves: close the main before the basement floods, and say why.

Example: checking a pharmacy's patient-flagging rules

A pharmacy system flags patients for review. quorum could check those rules, the way it checks a pump's.

RULE
"Flag for kidney dosing if clearance is under 30."

EXCEPTION
"...unless the patient is on dialysis."

MISSING DATA
"No lab result received yet."



✓ **Gaps and overlaps found before go-live**

? **"Assumed normal: no result received"**

⚑ **Every flag explained, for audits**

It would only flag patients for a pharmacist to review, never decide treatment. Patient data stays inside the hospital.

HONEST LIMITS

What quorum is not, and what it has not done yet

Saying this up front is part of what makes the rest believable.

It does not predict the future

It applies agreed rules to what it sees now. "Heart attack in 48 hours" needs a trained, proven model.

Not certified for safety-critical use

Medical devices, cars or anything that could hurt someone would need formal certification first.

Learning is not proven

Pre-registered tests showed real average gains, but not enough to clear every bar set in advance.

AI drafts still need a person

On real documents the large AI now gets the rules right in most runs, but not all. An engineer must approve.

It advises and decides inside a rulebook a person wrote and approved. It is a careful judge, not a fortune teller.

WHERE IT STANDS

Built and tested

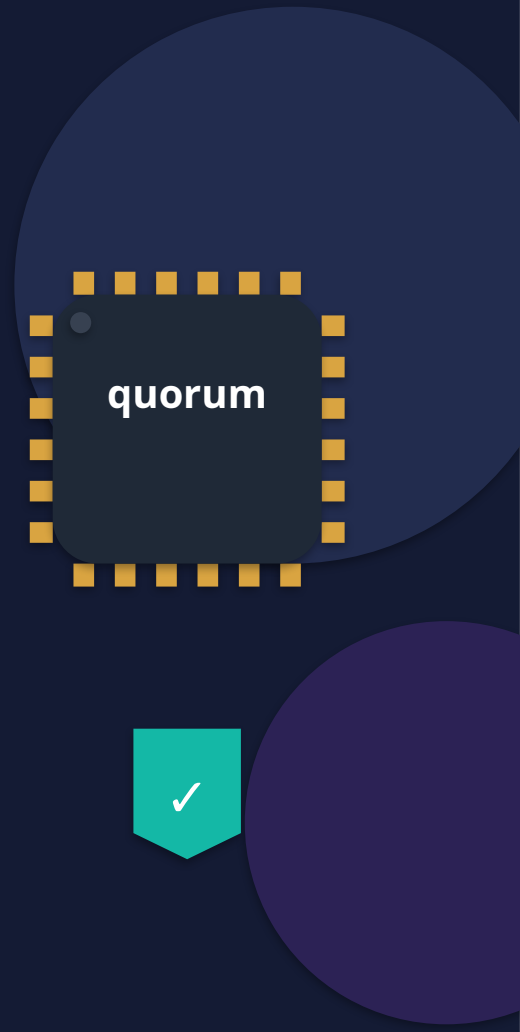
October 2026.

✓	**Engine and checker**	Every-case checking, refusals, explanations, speed limit.
✓	**Desktop studio**	Edit, import worksheets, learn from records, write chip files. Windows installer.
✓	**Three boards**	Pico 2 and Uno R4 match the PC exactly; Pi 5 runs everything natively.
✓	**AI guard, Modbus, proofs**	Checked on every situation; guard and Modbus reproduced on both chips.
→	**AI drafting**	Large AI models draft rulebooks from manuals; quorum checks them; a person approves.

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IN SUMMARY

A careful, honest judge for machines and their rules.



- Rules in plain words, checked against every situation before they run.
- One clear answer, on time, with its reasons and its assumptions.
- Small enough for an $11 chip; refuses anything it cannot trust.
- Useful well beyond factories: lifts, rail, farms, food, water, energy, healthcare rules.

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